

MySQL Interview Questions (Theory + Practical)

Basic MySQL Theory Questions

No.	Question
1	What is MySQL?
2	Difference Between Database and DBMS
3	What is RDBMS?
4	What are Tables, Rows, and Columns?
5	What is Primary Key?
6	Difference Between Primary Key and Unique Key
7	What is Foreign Key?
8	Difference Between DELETE, DROP, and TRUNCATE
9	Difference Between CHAR and VARCHAR
10	Difference Between WHERE and HAVING
11	What is NULL?
12	What is AUTO_INCREMENT?
13	Difference Between INT and BIGINT
14	What are Constraints?
15	What is Normalization?
16	What is Denormalization?
17	What are Joins?
18	Difference Between INNER JOIN and LEFT JOIN
19	What is Index?
20	Clustered vs Non-Clustered Index

SQL Query Questions

No.	Question
21	Find Second Highest Salary
22	Find Nth Highest Salary
23	Find Duplicate Records
24	Remove Duplicate Records
25	Find Employees Earning More Than Average Salary
26	Highest Salary Per Department

27	Top 3 Salaries Per Department
28	Find Departments Having More Than 5 Employees
29	Find Employees Without Department
30	Find Customers Without Orders
31	Find Employees Working Under Same Manager
32	Find Employees Earning More Than Their Manager
33	Find Department With Highest Average Salary
34	Find Running Total of Salaries
35	Rank Employees by Salary
36	Find Previous Row Salary
37	Find Next Row Salary
38	Find Consecutive Login Days
39	Find Missing IDs
40	Find Median Salary

Transaction & ACID Questions

No.	Question
41	What is Transaction?
42	What is ACID Property?
43	Difference Between COMMIT and ROLLBACK
44	What is SAVEPOINT?
45	What is Deadlock?
46	What are Isolation Levels?
47	Which Storage Engine Supports Transactions?

Performance Questions

No.	Question
48	How Does Index Improve Performance?
49	Why Query Becomes Slow?
50	What is EXPLAIN in MySQL?
51	What is Full Table Scan?
52	What is Composite Index?
53	Difference Between WHERE and INDEX Scan

CTE & Window Function Questions

No.	Question
54	What is WITH Clause?
55	What is CTE?
56	Difference Between CTE and Subquery
57	What is ROW_NUMBER()?
58	Difference Between ROW_NUMBER, RANK, and DENSE_RANK
59	What is PARTITION BY?
60	Difference Between GROUP BY and Window Functions

Practical Scenario Questions

No.	Question
61	Design Student Management Database
62	Design E-Commerce Database
63	How Will You Store Employee-Manager Relationship?
64	How to Handle Millions of Records Efficiently?
65	How to Improve Slow Query?

Python/Java + MySQL Questions

No.	Question
66	How to Connect Python with MySQL?
67	Which Python Library Used for MySQL?
68	Difference Between fetchone(), fetchall(), fetchmany()
69	Why commit() is Required?
70	What is Cursor in Python MySQL?
71	How to Prevent SQL Injection?

Advanced Interview Questions

No.	Question
72	Difference Between MyISAM and InnoDB

73	What Happens Internally When SELECT Query Runs?
74	Execution Order of SQL Query
75	Why Cannot We Use Aggregate Function in WHERE?
76	Difference Between UNION and UNION ALL
77	Difference Between EXISTS and IN
78	Correlated vs Non-Correlated Subquery
79	What is Self Join?
80	What is Cartesian Product?